Mathematical Expectation:

The *average* value of a random phenomenon is also termed as its ***mathematical expectation*** or ***expected value***.

The expected value of a discrete r.v. is a weighted average of all possible values of the r.v., where the weights are the probabilities associated with the corresponding values.

For a discrete r.v. $\boldsymbol{X}$ with p.m.f. $\boldsymbol{p(x)}$,

$$E(X) = \sum_x x\, p(x)$$

For a continuous r.v. $\boldsymbol{X}$ with p.d.f. $\boldsymbol{f(x)}$,

$$E(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx$$

Expected Value of a function of a random variable

Consider a r.v. X with *p.d.f. (p.m.f.)* $f(x)$ and distribution function $F(x)$.
If $g(\cdot)$ is a function such that g(X) is a r.v. and $E[g(X)]$ exists, then

$$E[g(X)] = \sum_x g(x)\, p(x) \qquad \text{(for discrete r.v.)}$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)\, f(x)\, dx \quad \text{(for continuous r.v.)}$$

Particular cases:

1. If we take $g(X) = X^r$, r - being a positive integer, then

$$E[X^r] = \int_{-\infty}^{\infty} x^r f(x)\ dx$$

$$E[X^r] = \sum_x x^r p(x)$$

Which is defined as $\boldsymbol{\mu_r'}$, the rth **moment (about origin)** of the probability distribution. Thus μ_r' (about origin) = $E(X^r)$

In particular, if $r = 1$ μ_1' (about origin) = $E(X)$ = $\boldsymbol{\bar{X}}$ = **Mean**

$r = 2$ μ_2' (about origin) = $E(X^2)$ = $\int_{-\infty}^{\infty} x^2 f(x)\ dx$

$$\mu_2 = \mu_2' - {\mu_1'}^2 = E(X^2) - [E(X)]^2$$

2. If $g(X) = E[X - E(X)]^r = E[X - \bar{x}\,]^r$, then

$$E[X - E(X)]^r = \int_{-\infty}^{\infty} [x - E(X)]^r f(x)\ dx = \int_{-\infty}^{\infty} [x - \bar{x}\,]^r f(x)\ dx$$

$$E[X - E(X)]^r = \sum_x [x - E(X)]^r p(x) = \sum_x [x - \bar{x}\,]^r p(x)$$

which is $\boldsymbol{\mu_r}$, the ***r*****th moment about mean**.

if $r = 2$, we get $\mu_2 = E[X - E(X)]^2 = \int_{-\infty}^{\infty} [x - \bar{x}\,]^2 f(x)\ dx$ = **Variance**

3. If $g(x)$ = constant = c,

$$E(c) = \int_{-\infty}^{\infty} c\, f(x)\ dx = c \int_{-\infty}^{\infty} f(x)\ dx = c$$

$$E(c) = \sum_x c\, p(x) = c \sum_x p(x) = c$$

Expected Value of Two-dimensional RV

If (X, Y) is a two-dimensional RV with joint *p.d.f.* $f(x, y)$ or *joint p.m.f.* p_{ij}, then

$$E[g(X,Y)] = \sum_i \sum_j g\left(x_i, y_j\right) p_{ij} \quad \text{(for discrete r.v.)}$$

$$E[g(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f(x,y) \; dxdy \quad \text{(for continuous r.v.)}$$

Properties of Expectation:

1. **Addition theorem of Expectation**: $\boldsymbol{E(X+Y) = E(X) + E(Y)}$

Proof: $E(X+Y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x+y)f(x,y)\ dxdy$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x)f(x,y)\ dxdy + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(y)f(x,y)\ dxdy$$

$$E(X+Y) = E(X) + E(Y)$$

Generalisation: $E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n)$

or $E(\sum_{i=1}^{n} X_i) = \sum_{i=1}^{n} E(X_i)$

2. Multiplication theorem of Expectation:

If X and Y are independent random variables, then $\boldsymbol{E(X\,Y) = E(X)\cdot E(Y)}$

$$E(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(xy)f(x,y)\ dxdy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(xy)f(x)\,f(y)dxdy$$ [Since, X and Y are independent]

$$= [\int_{-\infty}^{\infty} x\,f(x)\ dx]\ [\int_{-\infty}^{\infty} y\,f(y)\ dy]$$

$$E(X\,Y) = E(X)\,E(Y)$$

Generalisation: If $X_1, X_2, \dots, X_n$ are n- independent r.v.'s, then

$$E(X_1 \cdot X_2 \cdot \dots \cdot X_n) = E(X_1)\cdot E(X_2)\cdot \dots \cdot E(X_n)$$

or $$E(\prod_{i=1}^{n} X_i) = \prod_{i=1}^{n} E(X_i)$$

3. If $\boldsymbol{X}$ is a random variable and $\boldsymbol{a}$ and $\boldsymbol{b}$ are constants, then

$$\boldsymbol{E(aX+b) = a\,E(X)+b}$$

Proof: $E(aX+b) = \int_{-\infty}^{\infty}(ax+b)\,f(x)\,dx$

$$= a\int_{-\infty}^{\infty} x\,f(x)\,dx + b\int_{-\infty}^{\infty} f(x)\,dx$$

$$E(aX+b) = a\,E(X)+b$$

If $b = 0$, then we get $E(aX) = a\,E(X)$

However, $E\left(\frac{1}{X}\right) \neq \frac{1}{E(X)}$

$$E\left(X^{1/2}\right) \neq [E(X)]^{1/2}$$

$$E(\log X) \neq \log E(X)$$

$$E(X^2) \neq [E(X)]^2$$

Variance:

Variance = $\boldsymbol{Var(X)} = E[X - E(X)]^2 = E(X^2) - [E(X)]^2$

1. If X is a random variable, then $V(ax + b) = a^2 V(X)$

Proof: Let $Y = aX + b$

$$\mathrm{E}(Y) = E(aX + b)$$
$$= a\,E(X) + b$$
$$Y - E(Y) = (aX + b) - (a\,E(X) + b)$$
$$= a[X - E(X)]$$
$$\mathrm{E}[Y - E(Y)]^2 = a^2\,E[X - E(X)]^2$$
$$V(Y) = a^2 V(X)$$

2. $V(b) = 0$, *Variance of a constant is zero.*

3. $\sigma = \sqrt{Variance} = \sqrt{Var(X)}$ = Standard Deviation.

Example 1

Let X be a random variable with the following probability distribution:

X	:	-3	6	9
$P(X = x)$	:	1/6	1/2	1/3

Find the values of $E(X)$, $E(X^2)$ and $E(2X + 1)^2$

Example 2

Find the expectation of the number on a die when thrown.

Example 3:

Two unbiased dice are thrown. Find the expected values of the sum of numbers of points on them.

Example 4:

A coin is tossed until a head appears. What is the expectation of the number of tosses required.

Co-Variance:

If X and Y are two random variables, then the covariance between them is defined as:

$$Cov(X,Y) = E[\{X - E(X)\}\{Y - E(Y)\}]$$

$$= E[XY - X\,E(Y) - Y\,E(X) + E(X)E(Y)]$$

$$= \mathrm{E}(XY) - E(X)\,E(Y) - E(Y)\,E(X) + E(X)E(Y)$$

$$Cov(X,Y) \quad = \mathrm{E}(XY) - E(X)\,E(Y)$$

If X and Y are independent random variables, $\mathrm{E}(XY) = E(X)\,E(Y)$

and hence, $Cov(X,Y) \quad = E(X)\,E(Y) - E(X)\,E(Y) = 0$

Properties:

1. $Cov(aX, bY) = E[\{aX - E(aX)\}\{bY - E(bY)\}]$

   $= E[a\{X - E(X)\}\, b\{Y - E(Y)\}]$

   $= ab\, E[\{X - E(X)\}\, \{Y - E(Y)\}]$

   $Cov(X, Y) \quad = ab\; Cov(X, Y)$

2. $Cov(X + a, Y + b) = Cov(X, Y)$

3. $Cov\left(\frac{X - \bar{X}}{\sigma_X}, \frac{Y - \bar{Y}}{\sigma_Y}\right) = \frac{1}{\sigma_X\, \sigma_Y}\, Cov(X, Y)$

4. If X and Y are independent, $Cov(X, Y) = 0$, but the converse is not true.

5. $V(X_1 \pm X_2) = V(X_1) + \mathrm{V}(X_2) \pm 2Cov(X_1, X_2)$

6. $V(\sum_{i=1}^{n} a_i X_i) = \sum_{i=1}^{n} a_i^2\, V(X_i) + 2\, \sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j\, Cov(X_i, X_j)$

Correlation Coefficient:

If X and Y are two random variables, then the covariance between them is denoted as $Cov(X,Y)$,

$$Cov(X,Y) = \mathrm{E}(XY) - E(X)\,E(Y)$$

and the Co-efficient of correlation$(r_{XY} \;\; or \; \rho_{XY})$ between X and Y is defined as ***a numerical measure of linear relationship between them***,

$$\rho_{XY} = \frac{Cov(X,Y)}{\sigma_X \, \sigma_Y} = \frac{Cov(X,Y)}{\sqrt{Var(x)}\,\sqrt{Var(y)}}$$

Properties:

1. If X and Y are independent, $Cov(X, Y) = 0$ and $\rho_{XY} = 0$

   but the converse is not true.

   When $\rho_{XY} = 0$, we say that X and Y are uncorrelated.

2. $|\rho_{XY}| \leq 1$

   i.e., if $\rho_{XY} = +1 \Rightarrow$ Positive Correlation

   $\rho_{XY} = -1 \Rightarrow$ Negative Correlation

   $\rho_{XY} = 0 \Rightarrow$ UnCorrelated random variables.

Example 5: For a bivariate probability distribution of (X, Y) given below,

Y \ X	**-1**	**0**	**1**
-1	0	0.1	0.1
0	0.2	0.2	0.2
1	0	0.1	0.1

Find (i) Show that X and Y are independent.

(ii) Find $Var\ (X)$ and $Var(Y)$.

(iii) Find $r(X, Y)$.

Example 6

Two random variables X and Y have the following joint probability density function:

$$f(x,y) = \begin{cases} 2 - x - y\,;\ 0 \le x \le 1\ ,0 \le y \le 1 \\ \quad 0\ ; \qquad\qquad otherwise \end{cases}$$

Find
(i) Marginal probability density function of X and Y.
(ii) Conditional density function of X and Y
(iii) Var (X), Var (Y) and Corr(X,Y).